

Keuntae Kim

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Summary

PhD Candidate at Hanyang University researching how large language and multimodal models **reason, adapt continually, and remain grounded in evidence**. My primary focus is reasoning in LLMs and VLMs — including reinforcement learning and policy optimization, continual / incremental learning, and retrieval-augmented and meta-prompted reasoning. I extend this line of work to diffusion language and multimodal models (d(M)LLMs), Video-LLMs, Vision-Language-Action (VLA) models, and world models. To date, this has produced **3 first-author papers** accepted at **CVPR / ICML / NeurIPS 2025–2026** and **5+** additional first-author submissions under review at NeurIPS / EMNLP 2026, shipped as reproducible training and inference pipelines on PyTorch, DeepSpeed, and vLLM. I serve as **Lead Researcher** on two Korean national AI projects (NRF, IITP) on generative AI, and as a reviewer for NeurIPS, ICML, CVPR, ECCV, and ACL ARR.

Selected Projects

CLAWS — Creativity Detection for LLM-Generated Math Solutions *NeurIPS 2025*  github.com/kkt94/CLAWS

- Proposed **CLAWS**, a white-box detector that uses **attention weights across prompt sections vs. output** to classify LLM solutions into *Typical / Creative / Hallucinated* — the first three-way creativity assessment for reasoning, with no human evaluation in the loop.
- Validated on **5 7–8B math RL models** (DeepSeek-Math, Qwen-Math, Mathstral, OpenMath2-Llama3.1, Oreal) over **4,545 problems from 181 contests** (AJHSME, AMC, AIME).
- **Outperforms 5 white-box baselines** (Perplexity, Logit/Window Entropy, Hidden Score, Attention Score) on both creativity and hallucination detection.

Thinking Diffusion — Visual-Grounded Reasoning for Diffusion MLLMs

CVPR 2026

- Conducted the **first quantitative analysis of CoT reasoning in dMLLMs**, identifying two failure modes: *Early Answer Generation* (model commits to the answer before reasoning) and *weak early-step visual grounding*.
- Proposed two training-free inference-time methods — **Position & Step Penalty (PSP)** and **Visual Reasoning Guidance (VRG)** — that delay premature answers and amplify visual conditioning via classifier-free guidance.
- Across **LaViDa** and **MMaDA** on *M3CoT*, *MMBench*, *ScienceQA-IMG*, *V* Bench*: **up to +7.5%** accuracy and **>3× faster** inference vs. baselines using 4× more diffusion steps. Plug-and-play across dMLLMs with no retraining.

ST-Veto — Spatio-Temporal Token Veto for Diffusion MLLMs

ICML 2026

- Designed a training-free, decoding-time **veto-and-swap mechanism** that exploits a property unique to dMLLMs: visibility of all tokens across diffusion steps.
- Vetoes (i) **temporally unstable tokens** via second-order Taylor prediction of confidence dynamics and (ii) **weakly-grounded tokens** via image-attention mass, then swaps them for safer candidates before unmasking.
- On **LaViDa-reason** and **MMaDA-MixCoT** across *M3CoT*, *MMBench*, *ScienceQA-IMG*, *V* Bench* under long generation lengths: **up to +9%** accuracy over baseline decoding and prior VLM reasoning methods, with **zero extra training or generation cost**.

CWM — Controllable White-Box Meta-Prompting for Adaptive RAG & Reasoning *EMNLP ARR (March 2026)*

- Designed **CWM**, a low-cost white-box meta-prompting method that decides whether to retrieve using **model-internal signals only** — no external classifier, no multi-sampling.
- Achieves **state-of-the-art on 3 adaptive RAG benchmarks** across **GPT-oss-20b**, **Qwen3-14b**, **Llama3.1-8b**, while also generalizing to pure reasoning tasks (CoT / Plan-and-Solve / Self-Discover regimes).
- Exposes a tunable internal signal that lets users **control retrieval frequency at inference time** without retraining.

- Proposed a **Mixture-of-Experts projector** (Local Expert + State Expert, both Gated DeltaNet) that augments any frozen Video-LLM’s base projector to explicitly model inter-frame motion and global state transitions.
- Introduced **JEPA-style auxiliary objectives** — local correspondence + masked state reconstruction — to specialize each expert without touching the vision encoder or LLM.
- Trained only an **~80K-parameter** module for a **single epoch on 70K public samples**, yielding **up to ~7%** gains across **7 Video-LLMs** (PLM, VideoLLaMA3, InternVL3 families) on **5 benchmarks** (MVBench, TempCompass, NExT-QA, WorldBench, Perception Test).

Education

Hanyang University, Ph.D. in Computer Science

Sep. 2023 – Present

Advisor: Prof. Yong Suk Choi | Seoul, Korea

Hanyang University, M.S. in Computer Science

Sep. 2021 – Aug. 2023

Advisor: Prof. Yong Suk Choi | Seoul, Korea

Kookmin University, B.S. in Automobile and IT Convergence

Mar. 2015 – Aug. 2019

Seoul, Korea

Research Interests

- Diffusion Large/Multimodal Language Models (dLLMs, dMLLMs); visual-grounded reasoning
- Video-LLMs, World Models, Vision-Language-Action models
- Continual / Incremental Learning (stability-plasticity)
- Reinforcement Learning with Verifiable Rewards (RLVR); policy optimization for reasoning

Publications (Journals & Conferences)

Bold + underline = author; * = first author.

1. **Keuntae Kim***, Yong Suk Choi. “CFP-AL: Combining Model Features and Prediction for Active Learning in Sentence Classification.” *Applied Sciences*, 2025. **PUBLISHED**
2. **Keuntae Kim***, Eunhye Jeong, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. “CLAWS: Creativity detection for LLM-generated solutions using Attention Window of Sections.” *NeurIPS 2025*. **PUBLISHED**
3. **Keuntae Kim***, Mingyu Kang, Yong Suk Choi. “Thinking Diffusion: Penalize and Guide Visual-Grounded Reasoning in Diffusion Multimodal Language Models.” *CVPR 2026*. **PUBLISHED**
4. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. “ST-Veto: Spatio-Temporal Token Veto for Diffusion MLLMs via Taylor Prediction and Visual Grounding.” *ICML 2026*. **PUBLISHED**
5. **Keuntae Kim***, Eunhye Jung, Sehyeon Lee, Seohee Yoon, Yong Suk Choi. “Analyzing Creative Capabilities in RLVR Models: Overconfidence and High-Entropy Segments.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
6. **Keuntae Kim***, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi. “Revisiting Incremental Learning: A Three-Interface Diagnosis of Stability and Plasticity.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
7. **Keuntae Kim***, Jeonghyun Kim, Dongjun Kim, HyeongSeok Kang, Yong Suk Choi. “Projector-JEPA: A Gated DeltaNet MoE Projector Bridging Video-LLMs to World Understanding.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
8. JungHyoun Kim, **Keuntae Kim**, HyeongSeok Kang, Dongjun Kim, Yong Suk Choi. “FourDelta: Fourier-Band Delta Predictor for Latent World Models.” *NeurIPS 2026 (under review)*. **UNDER REVIEW**
9. **Keuntae Kim***, Eunhye Jeong, Yong Suk Choi. “CWM: Controllable White-Box Meta-Prompting for Adaptive Retrieval-Augmented Generation and Reasoning Ability.” *EMNLP 2026 (under review)*. **UNDER REVIEW**

Research Grants & Funded Projects

- **Metacognitive AGI Framework and its Applications** — IITP, Korea. *Lead Researcher* (Jul. 2025 – Present).
- **Metacognitive Generative AI Framework for Mitigating Hallucination in LLMs** — NRF, Korea. *Lead Researcher* (Feb. 2025 – Present).
- **Big Data-based Real-time Monitoring and Social Simulation System** — NRF, Korea. *Lead Researcher*,

Group 3 (Sep. 2021 – Feb. 2025).

- **High-performance Deep Learning via Cross-connection of Autoencoders and Decoders** — NRF, Korea. *Researcher* (Sep. 2021 – Feb. 2024).
- **Natural Language Data Refinement for Large-scale LM Construction** — KT Corp. industrial collaboration. *Researcher* (Sep. 2021 – Jul. 2022).

Patents (Korea)

- “Method and System for Fusion of Sparse-Dense and Global Attention for Video Action Recognition,” KR 10-2024-0132801 — *spatio-temporal attention fusion for video understanding*.
- “Device and Method for Correcting Toxic Words in Sentence,” KR 10-2024-0143175 — *LLM-based safety post-processing*.
- “Method and Device for Displaying Emotion Map Animation,” KR 10-2024-0152245 — *interactive web visualization of emotion trajectories over geo/time data*.
- “Method for Analysing Emotion Based on SNS Data,” KR 10-2023-0193546 — *end-to-end SNS crawling, NLP-based emotion analysis, and web-served dashboard*.

Technical Skills

Research areas. LLMs, VLMs, d(M)LLMs, Video-LLMs, World Models, RLVR, RAG, Continual Learning

Training. PyTorch, DeepSpeed, HF Accelerate, PEFT/LoRA, TRL, Megatron-LM, bitsandbytes, MS-swift

Serving & inference. vLLM, SGLang, TensorRT-LLM, HF TGI, ONNX Runtime, llama.cpp

Languages. Python, C/C++, CUDA, JavaScript, TypeScript, Java, SQL

Web & Backend. Flask, FastAPI, Django, Node.js, React, REST APIs, WebSocket

Infra. Linux, Docker, Git, W&B, MLflow, SLURM / multi-node GPU clusters

Awards & Honors

- **NeurIPS 2025 Scholar Award**
- **BK21 FOUR Scholarship** — Korean government-funded research fellowship, 2021 – Present

Academic Service

Conference & Journal Reviewer

- **CVPR 2026** — IEEE/CVF Conference on Computer Vision and Pattern Recognition
- **ECCV 2026** — European Conference on Computer Vision
- **NeurIPS 2026** — Conference on Neural Information Processing Systems
- **ICML 2026** — International Conference on Machine Learning
- **ACL ARR January 2026** — ACL Rolling Review
- **ACL ARR March 2026** — ACL Rolling Review
- **ACL ARR May 2026** — ACL Rolling Review